Dr. J. Wilson

Associate Professor

Contact

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Academic Qualifications: M.Sc.,Ph.D.

S.No	Degree	CollegeandUniversity	Year	Subject	Percentage
1	B.Sc.	Arul Anandar College Madurai Kamaraj University Madurai	1991	Physics	79%
2	M.Sc.	St. Joseph's College Bharathidasan University Trichy	1994	Physics	68%
3	Ph.D.	Alagappa University Karaikudi.	2006	Polymer Science	Highly Commented

TeachingExperience:15Years			
Designation and Research Institution	**From**	**To**	**Duties and Responsibilities**
Associate Professor	2020	Till date	Teaching and research in the field of Bioelectronics and Biosensors
Assistant Professor	2008	2020	

Additional Responsibilities

1. Member – Incubation and technology transfer center- Alagappa University, Karaikudi.
2. Member – Stock verification officer, Alagappa University, Karaikudi.
3. Member – Squad team for examinations of affiliated colleges, Alagappa University, Karaikudi.
4. Coordinator, Village placement programme, Dept. of Bioelectronics and Biosensors,
5. University Representative – Distance education examinations, Alagappa University, Karaikudi.
6. Counsellor - M.Sc. Bioelectronics students, Alagappa University, Karaikudi.
7. Member - Time table committee, Dept. of Bioelectronics and Biosensors.
8. Deputy Director, USIC.

Areas of Research

Conducting polymers, Metal oxides, Carbon based materials, Biosensors, lithium batteries, Natural polymers and Metal organic frameworks

Research Supervision/Guidance

ProgramofStudy		Completed	Ongoing
Research	**Ph.D.**	**6**	**2**
	M.Phil.	**2**	**-**
Project	**PG**	**46**	**5**

Publications

International		National		Others
Journals	Conferences	Journals	Conferences	Books /Chapters /Monographs / Manuals
49	55	-	25	3

CumulativeImpact Factor(asperJCR) : 256.3
h-index : 21
i10index : 34
TotalCitations : 1896

FundedResearchProjects

CompletedProjects

S. No	Agency	Period		ProjectTitle	Budget (Rs. Inlakhs)
		From	To		
1	UGC	2009	2012	Application of RuO_2 nanoparticle Doped conducting polymer(Poly-Pyrrole-polyaniline) composite for DNAsensor	10.14
2	RUSA-Alagappa University	2009	2010	Doping of RuO_2 nanoparticle-chitin Polyaniline composite for DNA Sensors.	0.64
3	BRNS-DAE	2015	2018	EB irradiated Polypyrrole-Polyaniline nanotubes doped with bimetallic nanoparticles for Biomolecules sensing using microarray electrodes	14.75
4	DST-SERB	2015	2018	Irradiation of PPy-PEDOT nanotubes doped with bimetallic nanoparticles for biomolecules sensing using micro electrodes.	30.40
5	UGC	2015	2018	Fabrication and characterization of irradiated Polypyrrole-Polyaniline PPy-PANi nanotubes doped with bimetallic nanoparticles for biomolecules sensing	10.21

6	RUSA-Alagappa University	2019	2020	Translational Health Research for Human, Animal and Plant Systems (TBRP) - Advanced materials for sustainable energy and sensor applications	6.78

OngoingProjects

S. No	Agency	Period		ProjectTitle	Budget (Rs. Inlakhs)
		From	To		
1	RUSA 2.0-Alagappa University	2023	2024	TBRP - Advanced materials for sustainable energy and sensor applications	10.00

DistinctiveAchievements/ Awards

1. Nature India has published my paper doi:10.1038/nindia.2016.27 Published online 24 February 2016
2. Promising Researcher Award – 2022, Alagappa University

Eventsorganizedinleadingroles

Number of Seminars / Conferences / Workshops / Events organized:Organizingsecretaryin

1. National Seminar on Advancements in Bioelectronics and Biosensors, 19th & 20th March, 2009, Alagappa University, Karaikudi 630003, India

2. National Seminar on Frontiers in Nanomaterials and Biosensors 4th & 5th March, 2010, Alagappa University, Karaikudi 630003,India

3. National conference on Recent Advancements in Nanomaterials for Sensor Applications (NANOSE-12), 8th & 9th March, 2012, Alagappa University, Karaikudi 630003, India

4. Organized one day workshop on Metrohm Autolab Electrochemical Instruments for biosensor, energy and corrosion applications, 16-02-2015, Alagappa University, Karaikudi, India

5. Member of the Assessment Committee (CSIR-Central Electrochemical Research Institute), 2018.

6. Organizing secretary in conference Biosensors in Agricultural, Environmental and Medical Sciences (2017), 13-14 March 2018

7. Organizing secretary in conference NanoBiosensors: Present Status and Future Perspectives (Nanose-2018), 8-9 March 2018.

8. Organizing secretary in conference Nanomaterials Driven Advances in Chemical and Biosensors (NanoSe - 2019), 27-29, November 2019.

EventsParticipated

Conferences/Seminars/Workshops: 95

1. Conferences - 80
2. Seminars - 10
3. Workshops - 5

Other Training Programs

1. OrientationProgramme - 3
2. Refresher Course (FDP, short term course etc.) - 12

Overseas Exposure/Visits

1. Singapore
2. Malaysia

Membership in

Professional Bodies

1. Life Member: The Indian science congress Association
2. Life Member: Material research society of India (MRSI)

Academic Bodies (such as Board of Studies etc.,)

1. Member – Board of Studies, M.Sc. Electronics, Alagappa University, Karaikudi.

Resource persons in various capacities

Number of Invited Lectures delivered: 15

Others

No. of Ph.D. Public Viva Voce Examination conducted: 1

RecentPublications

1. Nathiya Dhananjayan, Wilson Jeyaraj, Gurunathan Karuppasamy, Mohana Rani Gokana, C Venkateswara Raju and Vishal Chaudhary; Ultrasensitive Detection of Bovine Serum Albumin on DNA Modified Protein Microcapsules-based Electrodes; Journal of The Electrochemical Society, 2023.

2. T. Thenrajan, Muthaiah Anandhakumar, Mohana Rani Gokana, Vishal Chaudhary, C Venkateswara Raju and Wilson Jeyaraj; Guar Gum Supported ZIF-8 as an Effective Catalyst for Electrochemical Sensing of Gallic Acid in Liquid Food Samples; Journal of the electrochemical society, 2023.

3. Thatchanamoorthy Thenrajan, Subbiah Alwarappan and Jeyaraj Wilson: Molecular Diagnosis and Cancer Prognosis — A Concise Review: Diagnostics, 13, 766, 2023.

4. S. Dheepthi GunaVathana, Shibasini Murugan, S. Girija, J. Wilson, Kavitha Thangavel, and A. Cyrac Peter; Palladium nanospheres incorporated polythiophene nanocomposite: Investigation of potency promising antimicrobial efficacy: Inorganic Chemistry Communications, 144, 10986, 2022.

5. S. Girija, S. Sam Sankar, Subrata Kundu and J Wilson; Selective Determination of Catechol Using One Dimensional Zeolitic Cobalt–Nickel Imidazolate Framework Journal of Inorganic and Organometallic Polymers and Materials, 32, 3837, 2022.

6. Thatchanamoorthy Thenrajan and Jeyaraj Wilson: Biosensors for cancer theranostics Biosensors and Bioelectronics: X, 100232, 2022.

7. Thatchanamoorthy Thenrajan, Sam Sankar Selvasundarasekar, Subrata Kundu and Jeyaraj Wilson; Novel Electrochemical Sensing of Catechins in Raw Green Tea Extract via a Trimetallic Zeolitic Imidazolate Fibrous Framework, ACS omega, 7, 19754, 2022.

8. S. Dheepthi Gunavathana, S. Girija, J. Wilson and A. Cyrac Peter; ZnO nanorods bonded polythiophene nanocomposite: an enhanced electrochemical voltammetric biosensing of L-tryptophan; Bulletin of Materials Science, 45, 57, 2022.

9. Thatchanamoorthy Thenrajan, Selvasundarsekar Sam Sankar, Subrata Kundu and Jeyaraj Wilson; Bimetallic nickel iron zeolitic imidazolate fibers as biosensing platform for neurotransmitter serotonin; Colloid and Polymer Science, 300223, 2022.

10. Thatchana Moorthy Thenrajan, Sriramprabha Ramasamy, Pandi Keerthika Chidambaram and Jeyaraj Wilson; A green approached biocomposite: Iron (III) oxide dissemination over cassava starch for selective detection of epinephrine; Materials Chemistry and Physics, 276, 125366 and 2022.

11. J. Wilson and D. Nathiya; Fabrication of nitrogen doped graphene decorated extended gate field effect transistor: Application in protein sensor; ACS Fall 2022.

12. Pannerselvam Thivya, Thatchanamoorthy Thenrajan, Rajendran Ramya, Jeyaraj Wilson; Polyaniline/sulfonated cassava starch: A green biocomposite for sensing of pyridoxine Current Research in Biotechnology, 4, 309, 2022

13. R. Ramya, D. Nathiya, P. Thivya and J. Wilson; Functionally anchored Ag-TiO2 nanoparticles on guar gum based nanocomposite for simultaneous determination of hydroquinone, catechol, resorcinol and nitrite; Microchemical Journal, 170, 106734, 2021.

14. Nathiya Dhananjayan, Karthika Viswanathan, Wilson Jeyaraj, Arumugam Ayyakannu and Gurunathan Karuppasamy; Antibiofilm and antimicrobial efficacy evaluation of polypyrrole nanotubes embedded in aminated gum acacia based nanocomposite; IET nanobiotechnology, 15, 441, 2021.

15. R. Ramya, P. Thivya, D. Nathiya and J. Wilson; Polypyrrole and enzyme free cholesterol flakes based composite: Selective determination of theophylline; Journal of Pharmaceutical and Biomedical Analysis, 199, 114065, 2021.

16. S. Dheepthi GunaVathana, J. Wilson, R. Prashanthi, A. Cyrac Peter; CuO nanoflakes anchored polythiophene nanocomposite: Voltammetric detection of L-Tryptophan; Inorganic Chemistry Communications, 124, 108398, 2021.

17. Thatchanamoorthy Thenrajan, Selvasundarsekar Sam Sankar, Girija Srinivasan, Subrata Kundu and Jeyaraj Wilson; Cobalt–iron zeolitic imidazolate frameworks (ZIFs) as microfibers for the effective detection of hydroquinone; Dalton Transactions, 50, 10540, 2021.

18. S. Girija, S. Sam Sankar, T. Thenrajan, Subrata Kundu and J. Wilson; Bi-metallic zeolite imidazole framework nanofibers for the selective determination of Cd 2+ ions; Journal of Materials Chemistry B, 9, 5656, 2021.

19. R. Sriramprabha, M. Sekar, R. Revathi, C. Viswanathan, J. Wilson; Fe_2O_3/polyaniline supramolecular nanocomposite: A receptor free sensor platform for the quantitative determination of serum creatinine; Analytica Chimica Acta, 1137, 103, 2020.

20. S. Girija, S. Sam Sankar, Subrata Kundu, J. Wilson ; Microfibers of Embellished Cobalt—Zeolite Imidazole Framework for Vitamin—B2 Detection; Journal of Electrochemical Society, 167, 137511, 2020.

21. P Thivya, R Ramya and J. Wilson; Poly (3, 4-ethylenedioxythiophene)/Taurine biocomposite on screen printed electrode: non-enzymatic Cholesterol biosensor; Microchemical Journal, 157, 105037, 2020.

22. S. Dheepthi GunaVathana, P. Thivya, J. Wilson and A. Cyrac PeterSensitive voltammetric sensor based on silver dendrites decorated polythiophene nanocomposite: Selective determination of L-Tryptophan; Journal of Molecular Structure; 1205, 127649 ,2020.

23. Dhananjayan Nathiya, Karuppasamy Gurunathan and Jeyaraj Wilson; Size controllable, pH triggered reduction of bovine serum albumin and its adsorption behavior with SnO_2/SnS_2 quantum dots for biosensing application; Talanta, 210, 120671, 2020.

24. G. Vinodhkumar, J. Wilson, S.S.R Inbanathan, I. Vetha Potheher, Muthupandian Ashok kumar and A. Cyrac Peter; Solvothermal synthesis of magnetically separable reduced graphene oxide/Fe_3O_4 hybrid nanocomposites with enhanced photocatalytic properties; Physica B: Condensed Matter; 580; 411752, 2020.

25. R. Sriramprabha, M. Sekar, Wilson Jeyaraj, N. Ponpandian, C. Viswanathan; Mesoporous nickel oxide nanostructures: Influences of crystalline defects and morphological features on mediator free electrochemical monosaccharide sensor application; Nanotechnology, 31, 21, 2020.

26. P. Thivya, J. Wilson and R. Ramya; J. Environmental pollutants simultaneous determination: DNA catalyst mediated polyaniline biocomposite nanostructures; Biocatalysis and Agricultural Biotechnology, 21, 101352, 2019.

27. Wilson Jeyaraj, Gurunathan Karuppasamy, Nathiya Dhananjayan ; Interactive Studies on Synthetic Nanopolymer decorated with Edible Biopolymer and its Selective Electrochemical determination of L-Tyrosine; scientific reports, 9, 13287, 2019.

28. G. Vinodhkumar, J. Wilson, S.Mahalakshmia, V.Ragavendran and A. Cyrac Peter; One step solvothermal synthesis and characterization of rGO/NiO nanocomposites; Materials Today Proceedings, 2019

29. P. Thivya and J. Wilson; Electron rays irradiated polyaniline anchored over bovine serum albumin for simultaneous detection of epinephrine and uric acid; Microchemical Journal, 145, 883-891, 2019.

30. P. Muthukumaran, R. Ramya, P. Thivya, J. Wilson, G. Ravi; Nanocomposite based on restacked crystallites of β-NiS and Ppy for the determination of theophylline and uric acid on screen-printed electrodes; New Journal of Chemistry, 43, 19397-19407, 2019.

31. R. Ramya, P. Muthukumaran and J. Wilson; Electron beam-irradiated polypyrrole decorated with Bovine serum albumin pores: Simultaneous determination of epinephrine and L-tyrosine; Biosensors and Bioelectronics, 108, 53-61, 2018.

32. Nathiya Dhananjayan, Muthukumaran Palanisamy, Wilson Jeyaraj and Gurunathan Karuppasamy; Stable and robust nanobiocomposite preparation using aminated guar gum (mimic activity of graphene) with electron beam irradiated polypyrrole and Ce-Ni bimetal: Effective role in simultaneous sensing of environmental pollutants and pseudocapacitor applications; Electrochimica Acta, 246, 484-496, 2017.

33. Dhanasekaran Solairaj, Palanivel Rameshthangam, Palanisamy Muthukumaran and Jeyaraj Wilson; Studies on electrochemical glucose sensing, antimicrobial activity and cytotoxicity of fabricated copper nanoparticle immobilized chitin nanostructure; International journal of biological macromolecules, 101, 668-679, 2017.

34. J. Wilson* and G. Ravi P. Muthukumaran, C. Sumathi; Enzymeless biosensor based on b-NiS@rGO/Au nanocomposites for simultaneous detection of ascorbic acid, epinephrine and uric acid; RSC Advances, 6, 96467–96478, 2016.

35. C. Sumathi, C. Venkateswara Raju, P. Muthukumaran, J. Wilson and G. Ravi; Au–Pd bimetallic nanoparticles anchored on α-Fe_2O_3 nonenzymatic hybrid nanoelectrocatalyst for

simultaneous electrochemical detection of dopamine and uric acid in the presence of ascorbic acid; Journal of Materials Chemistry B, 4, 2561-2569, 2016.

36. C. Sumathi, P. Muthukumaran, P. Thivya, J. Wilson and G. Ravi; DNA mediated electrocatalytic enhancement of α-Fe_2O_3 –PEDOT–C-MoS_2 hybrid nanostructures for riboflavin detection on screen printed electrode; RSC Advances, 6, 81500-81509, 2016.

37. Sathish Rajendran, Muthukumaran P ,Chikkili Venkateswara Raju , Sumathi C , S, Ravi G , Solairaj D , Rameshthangam P , Wilson J and Subbiah Alwarappan; Cerium doped nickel oxide nanostructures for riboflavin biosensing and antibacterial applications; New Journal of Chemistry, 40, 2741-2748, 2016.

38. C. Sumathi, P. Muthukumaran, S. Radhakrishnan, G. Ravi and J. Wilson; Riboflavin detection by α-Fe_2O_3 /MWCNT/AuNPs-based composite and a study of the interaction of riboflavin with DNA; RSC Advances, 5, 17888-17896, 2015.

39. Sivaprakasam Radhakrishnan, Karthikeyan Krishnamoorthy, Chinnathambi Sekar, Jeyaraj Wilson and Sang Jae Kim; A promising electrochemical sensing platform based on ternary composite of polyaniline– Fe_2O_3–reduced graphene oxide for sensitive hydroquinone determination; Chemical Engineering Journal, 259, 594-602, 2015.

40. R. Kumar, MS Chauhan, GN Dar, SG Ansari, J Wilson, Ahmad Umar, S Chauhan, DS Rana and P. Sharma; ZnO nanoparticles: Efficient material for the detection of hazardous chemical; Sensor letters, 12, 1393-1398, 2014.

41. Sivaprakasam Radhakrishnan, Karthikeyan Krishnamoorthy, Chinnathambi Sekar, Jeyaraj Wilson and Sang Jae Kim; A highly sensitive electrochemical sensor for nitrite detection based on Fe_2O_3 nanoparticles decorated reduced graphene oxide nanosheets; Applied Catalysis B: Environmental, 148, 22-28, 2014.

42. C. Sumathi, P. Muthukumaran, S. Radhakrishnan, J. Wilson and Ahmad Umar; Controlled growth of single-crystalline nanostructured dendrites of α- Fe_2O_3 blended with MWCNT: a systematic investigation of highly selective determination of l-dopa; RSC Advances, 4, 23050-23057, 2014.

43. P. Muthukumaran, C. Sumathi, J. Wilson, C. Sekar, SG Leonardi and G. Neri; Fe_2O_3/Carbon nanotube-based resistive sensors for the selective ammonia gas sensing; Sensor letters , 12, 17-23, 2014

44. S. Radhakrishnan, C. Sumathi, Ahmad Umar, Sang Jae Kim, J Wilson and V. Dharuman; Polypyrrole–poly (3, 4-ethylenedioxythiophene)–Ag (PPy–PEDOT–Ag) nanocomposite films for label-free electrochemical DNA sensing; Biosensors and Bioelectronics, 47, 133-140, 2013.

45. S. Radhakrishnan, C. Sumathi, V. Dharuman and J. Wilson; Gold nanoparticles functionalized poly (3, 4-ethylenedioxythiophene) thin film for highly sensitive label free DNA detection; Analytical Methods; 5, 684-689, 2013.

46. S. Radhakrishnan, C. Sumathi, V. Dharuman and J. Wilson; Polypyrrole nanotubes–polyaniline composite for DNA detection using methylene blue as intercalator; Analytical Methods, 5, 1010-1015, 2013.

47. J. Wilson, S. Radhakrishnan, C. Sumathi and V. Dharuman; Polypyrrole–polyaniline–Au (PPy–PANi–Au) nano composite films for label-free electrochemical DNA sensing; Sensors and Actuators B: Chemical, 171, 216-222, 2012.

48. V. Dharuman, K. Vijayaraj, S. Radhakrishnan, T. Dinakaran, J. Shankara Narayanan, M. Bhuvana and J. Wilson; Sensitive label-free electrochemical DNA hybridization detection in the presence of 11-mercaptoundecanoic acid on the thiolated single strand DNA and mercaptohexanol binary mixed monolayer surface; Electrochimica acta; 56, 8147-8155, 2011.

49. A. Manuel Stephan, Kee Suk Nahm, M. AnbuKulandainathan, G. Ravi and J. Wilson; Electrochemical studies on nanofiller incorporated poly (vinylidene fluoride–hexafluoropropylene) (PVdF–HFP) composite electrolytes for lithium batteries; Journal of applied electrochemistry, 36, 1091-1097, 2006.

50. A. Manuel Stephan, Kee Suk Nahm, T. Prem Kumar, M. AnbuKulandainathan, G. Ravi and J Wilson; Nanofiller incorporated poly (vinylidene fluoride–hexafluoropropylene)(PVdF–HFP) composite electrolytes for lithium batteries; Journal of power sources, 159 , 1316-1321, 2006.

51. A. Manuel Stephan, Kee Suk Nahm, M AnbuKulandainathan, G. Ravi and J. Wilson; Poly (vinylidene fluoride-hexafluoropropylene) (PVdF-HFP) based composite electrolytes for lithium batteries; European Polymer Journal, 42, 1728-1734, 2006.

52. J. Wilson, G. Ravi and M. AnbuKulandainathan; Electrochemical studies on inert filler incorporated poly (vinylidene fluoride-hexafluoropropylene) (PVDF-HFP) composite electrolytes; Polimeros, 16, 88-93, 2006.